

Deep Learning – Transfer Learning

CIML Summer Institute 2026

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DEEP LEARNING OVERVIEW

- Neural Network Basics
 - Processing Unit
 - Activation Function
 - Loss Function
 - Neural Network Training
- Deep Learning Fundamentals
 - Deep Network Layers
 - DL Architectures
 - DL Libraries
- Transfer Learning
 - Transfer Learning Concepts
 - Transfer Learning Demo

Transfer Learning

- To overcome challenges of training model from scratch:
 - Insufficient data
 - Very long training time
- Use pre-trained model
 - Trained on another dataset
 - This serves as starting point for model
 - Then train model on current dataset for current task

Transfer Learning Approaches

- Feature extraction
 - Remove classification layer from pre-trained model
 - Treat rest of network as feature extractor
 - Use features to train new classifier
 - “top model” or “classification head”
- Fine tuning
 - Tune weights in some layers of original model (along with weights of top model)
 - Train model for current task using new dataset

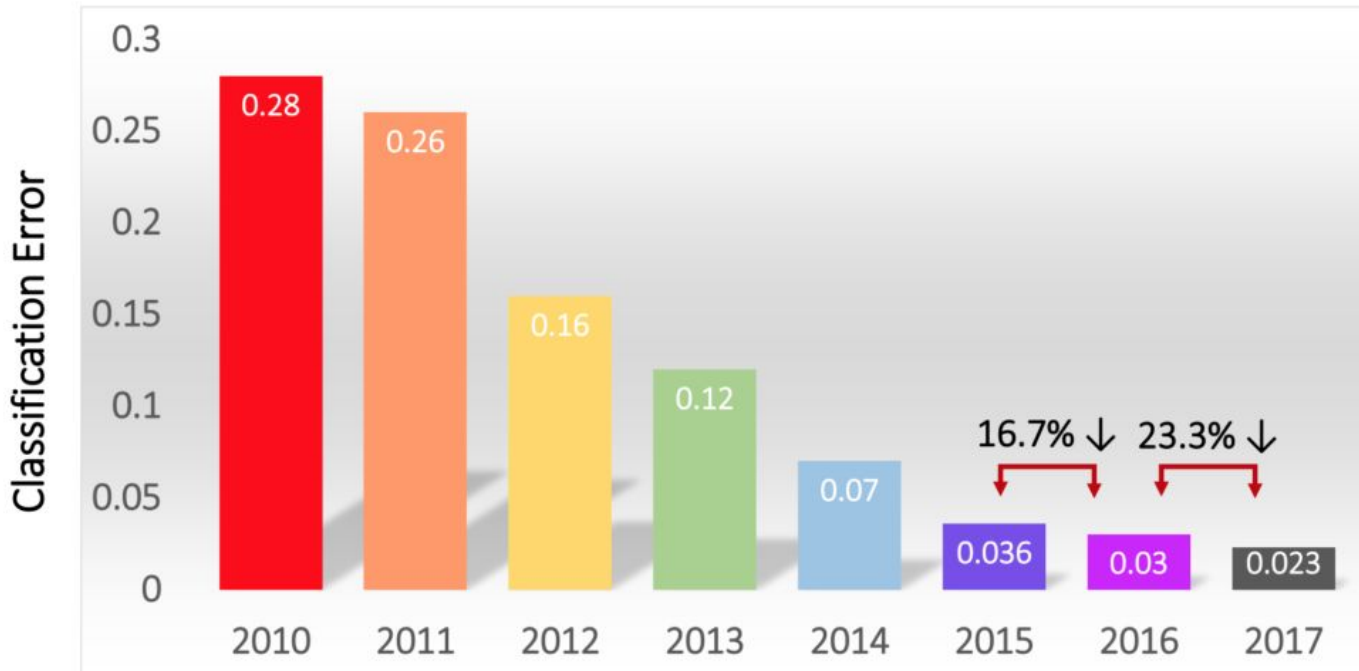
CNNs for Transfer Learning

- Popular architectures
 - AlexNet
 - GoogLeNet
 - VGGNet
 - ResNet
- All winners of ILSVRC
 - ImageNet Large Scale Visual Recognition Challenge
 - Annual competition on vision tasks on ImageNet data

- Database
 - Developed for computer vision research
 - ~14,000,000 images hand-annotated
 - ~22,000 categories
- ILSVRC History
 - Started in 2010
 - Image classification task: 1,000 object categories
 - Image classification error rate
 - 2010: 28.20% (conventional image processing techniques)
 - 2012: 15.30% (AlexNet)
 - 2015: 3.57% (ResNet; better than human performance)
 - 2016: 2.99% (16.7% error reduction)
 - 2017: 2.25% (23.3% error reduction)

Results on ImageNet Classification

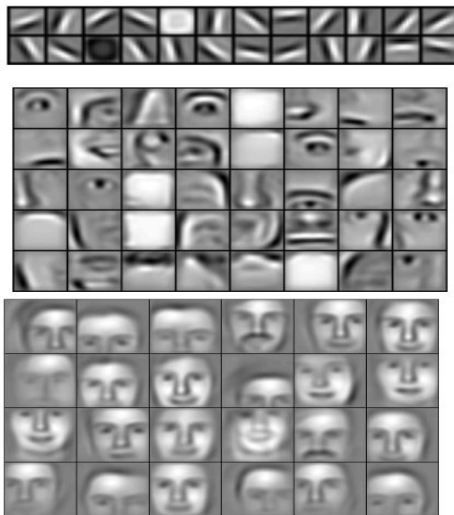
Classification Results (CLS)



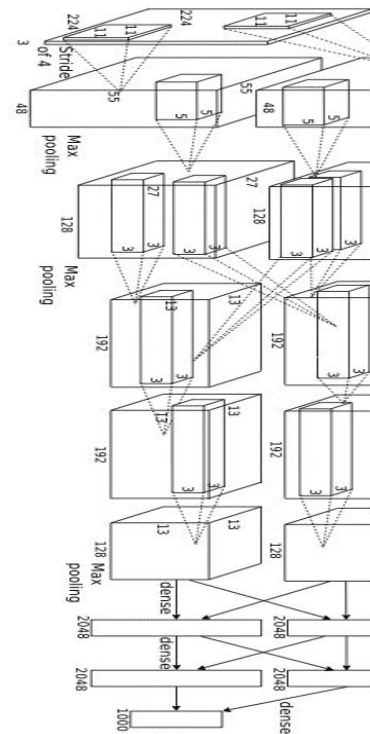
Transfer Learning

*Learned
hierarchy*

Input

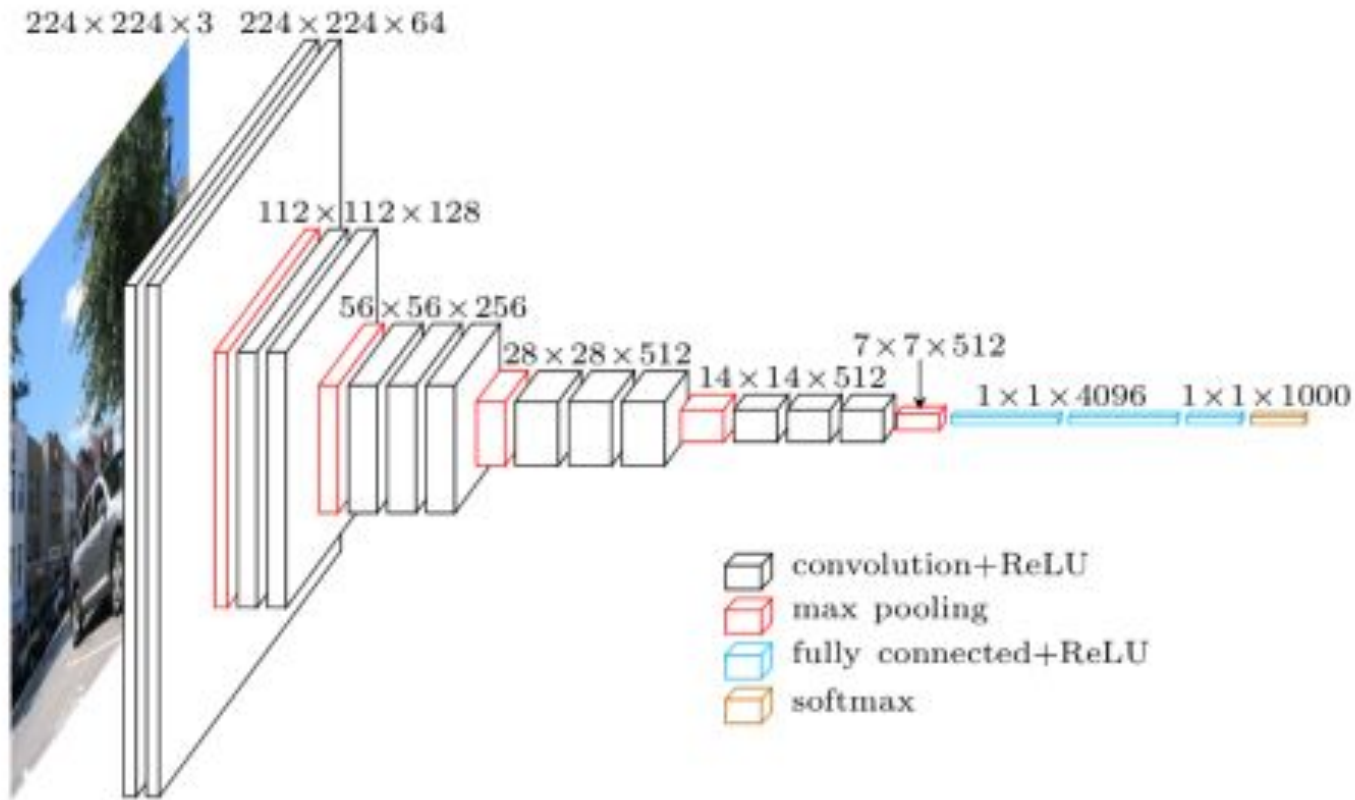


Output

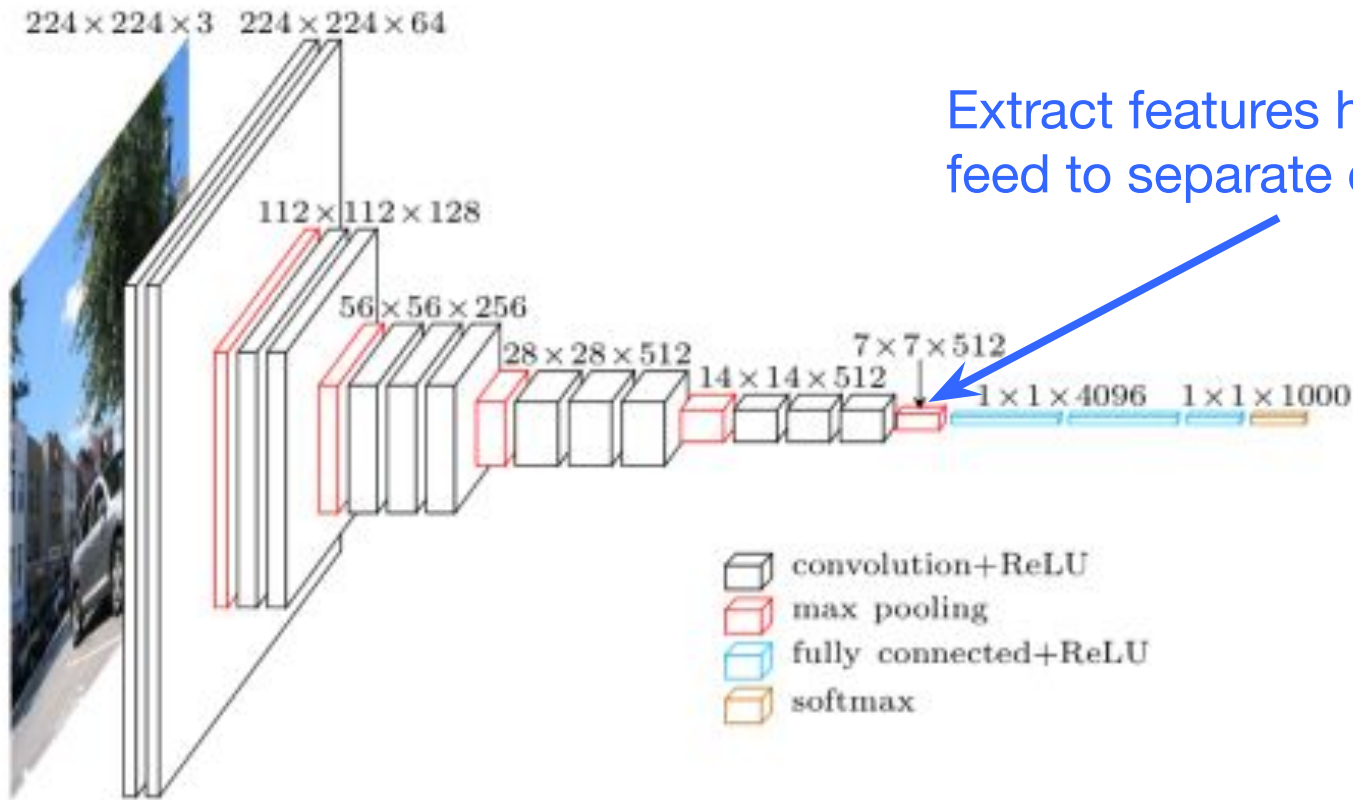


Lee et al. 'Convolutional Deep Belief Networks for Scalable
Unsupervised Learning of Hierarchical Representations' ICML 2009

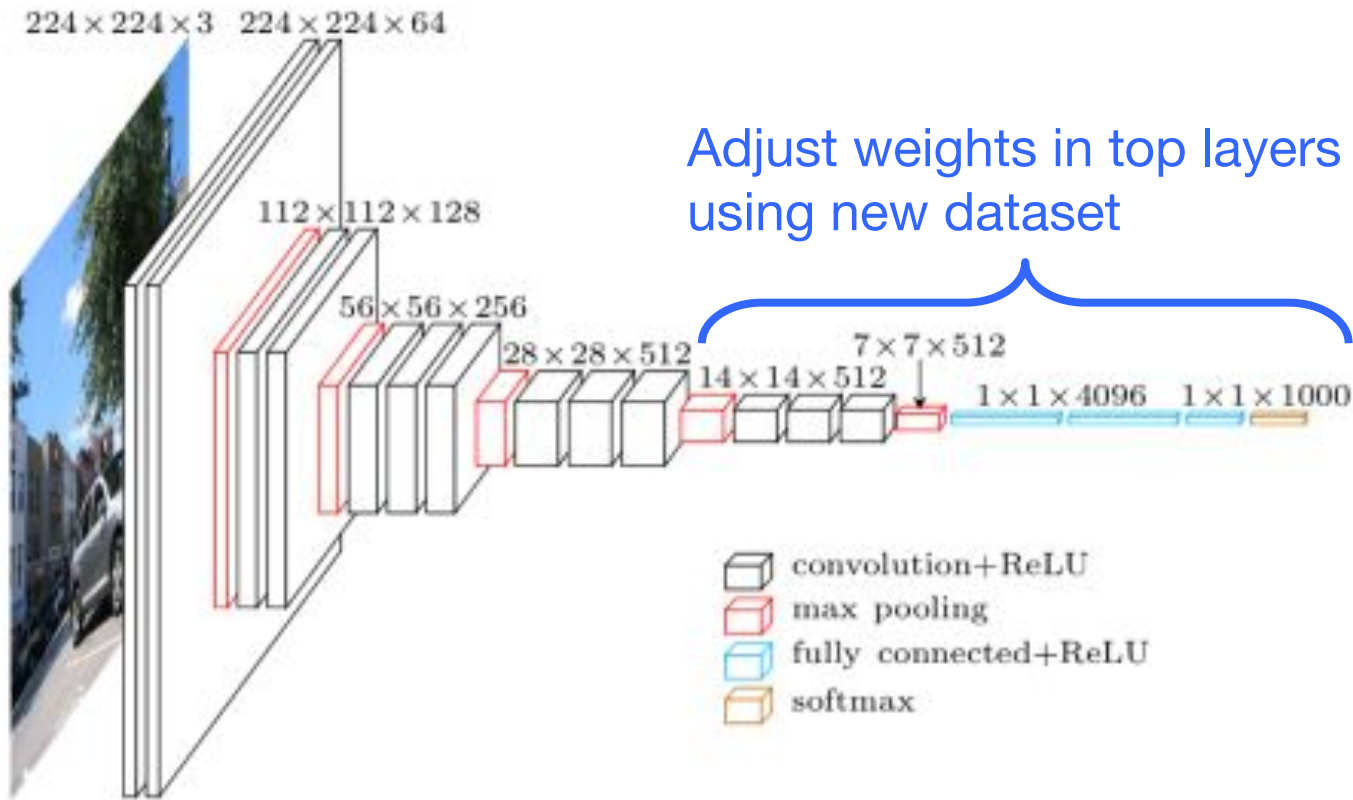
Pre-Trained Model



Transfer Learning - Feature Extraction



Transfer Learning - Finetuning



Practical Tips for Transfer Learning

- Learning rate
 - Use very small learning rate for fine tuning. Don't want to destroy what was already learned.
- Start with properly trained weights
 - Train top-level classifier first, then fine tune lower layers.
 - Top model with random weights may have negative effects on when fine tuning weights in pre-trained model
- Data augmentation
 - Simple ways to slightly alter images
 - Horizontal/vertical flips, random crops, translations, rotations, etc.
 - Use to artificially expand your dataset

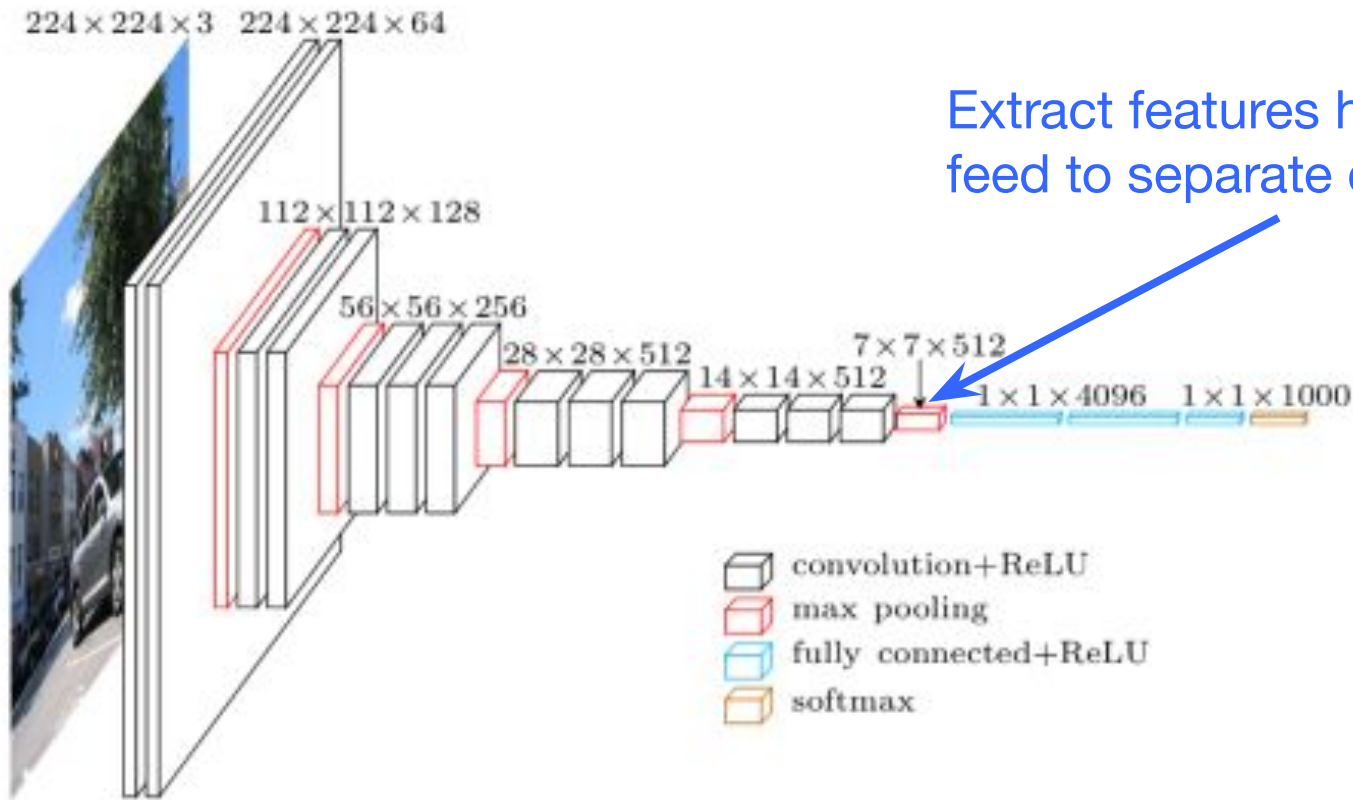
Transfer Learning Hands-On

- Data
 - Cats and dogs images from Kaggle
- Exercises
 - Feature extraction
 - Use pre-trained CNN to extract features from images
 - Train neural network to classify cats/dogs using extracted features
 - Fine tune
 - Adjust weights of last few layers of pre-trained CNN and top classifier model through training

- Subset of Dogs Vs. Cats dataset from Kaggle
 - <https://www.kaggle.com/c/dogs-vs-cats>
- Train
 - 1000 cats + 1000 dogs
- Validation
 - 200 cats + 200 dogs
- Test
 - 200 cats + 200 dogs



Transfer Learning - Feature Extraction



Feature Extraction Overview

- Code
 - `feature_extraction_hf.ipynb`
- Data
 - Set image dimensions & location
 - Read images from folder in batches
- Model
 - Load model pre-trained on ImageNet data
 - Freeze weights in pre-trained model to use as feature extractor
 - Add top model to classify cats vs dogs
 - Model = Pre-trained base model + top model classifier
- Train model
 - Use training data to adjust top model weights
- Evaluate model
 - Calculate accuracy, etc.
 - Perform inference on test images

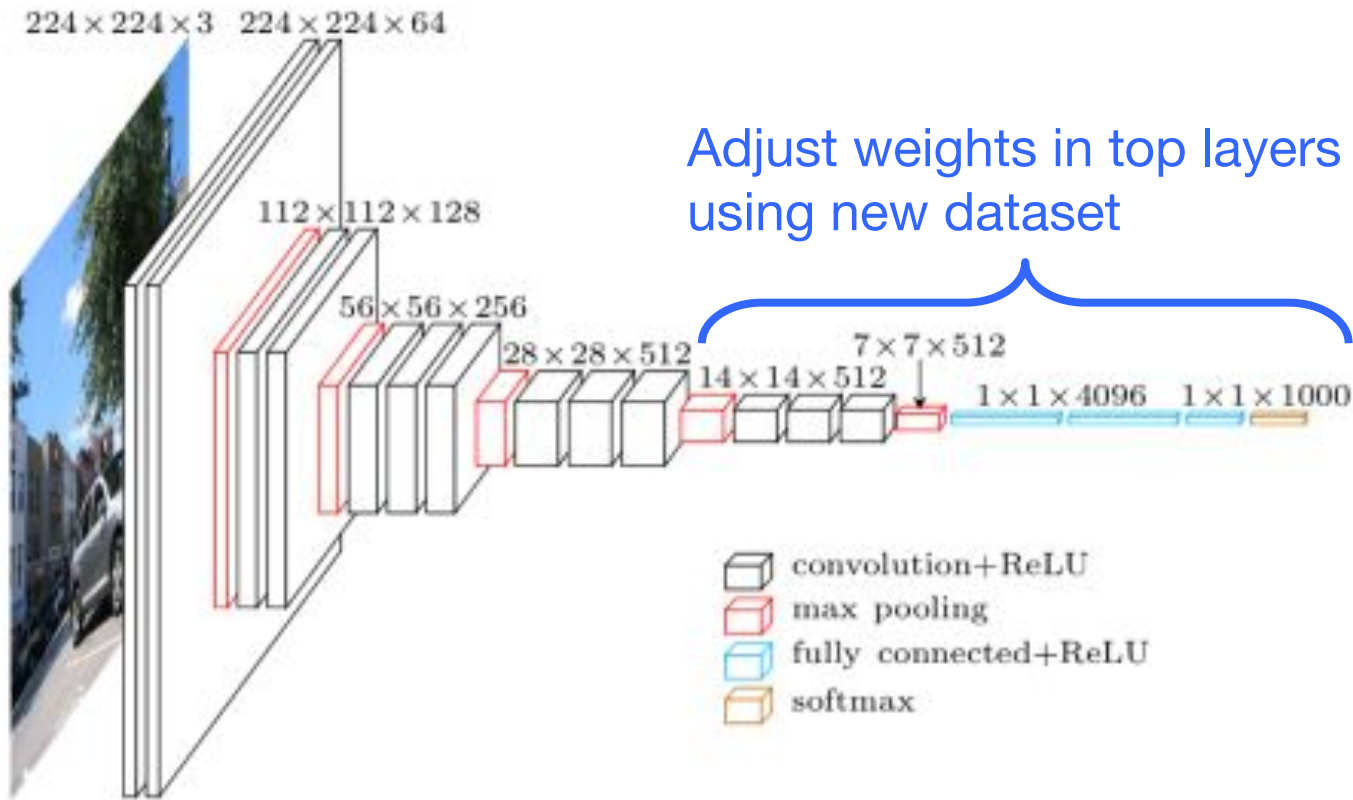
Server Setup on Expanse

- Login to Expanse through terminal window
 - Open terminal window on local machine and type in:
`ssh login.expanse.sdsc.edu -l <username>`
- Pull latest from repo
 - `git pull`
 - URL: <https://github.com/ciml-org/ciml-summer-institute-2026>
- In terminal window
 - `jupyter-nairr-gpu-shared-pytorch`
 - Alias for:
`galyleo launch --account ${CIML26_ACCOUNT} --reservation
${CIML26_RES_GPU} --partition nairr-gpu-shared --qos
${CIML26_QOS_GPU} --cpus 4 --memory 92 --gpus 1 --time-limit
04:00:00 --sif ${CIML26_DATA_DIR}/ptl-cuda-12-1.sif --bind
/expanse,/scratch,/cm --nv --quiet`
 - Copy and paste URL in local web browser
- To check queue
 - `squeue -u $USER`

Data

- In terminal window in Jupyter Lab, do the following
- Get counts of images
 - `ls`
 - `ls -l $CIML26_DATA_DIR/catsVsDogs/train/cats/* | wc -l`
 - `ls -l $CIML26_DATA_DIR/catsVsDogs/train/dogs/* | wc -l`
 - `ls -l $CIML26_DATA_DIR/catsVsDogs/val/cats/* | wc -l`
 - `ls -l $CIML26_DATA_DIR/catsVsDogs/val/dogs/* | wc -l`
 - `ls -l $CIML26_DATA_DIR/catsVsDogs/test/cats/* | wc -l`
 - `ls -l $CIML26_DATA_DIR/catsVsDogs/test/dogs/* | wc -l`

Transfer Learning - Finetuning

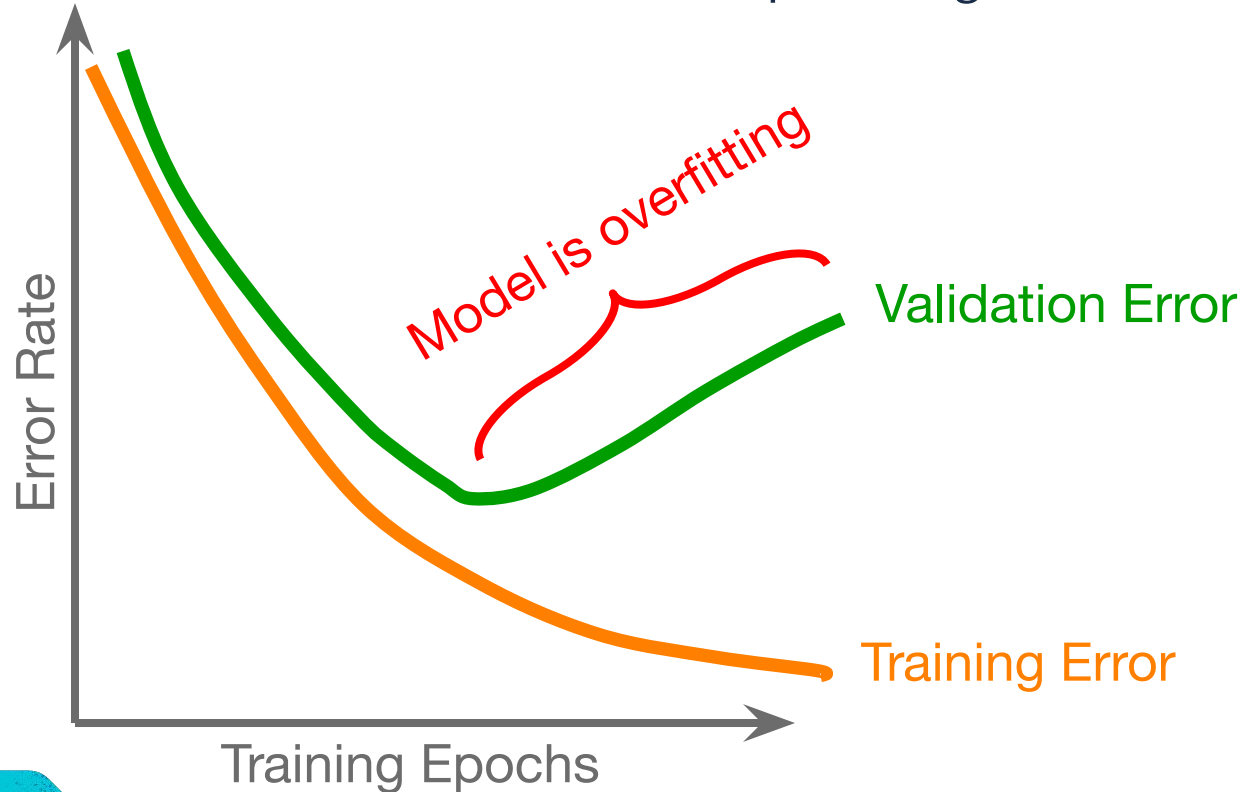


Fine Tune Overview

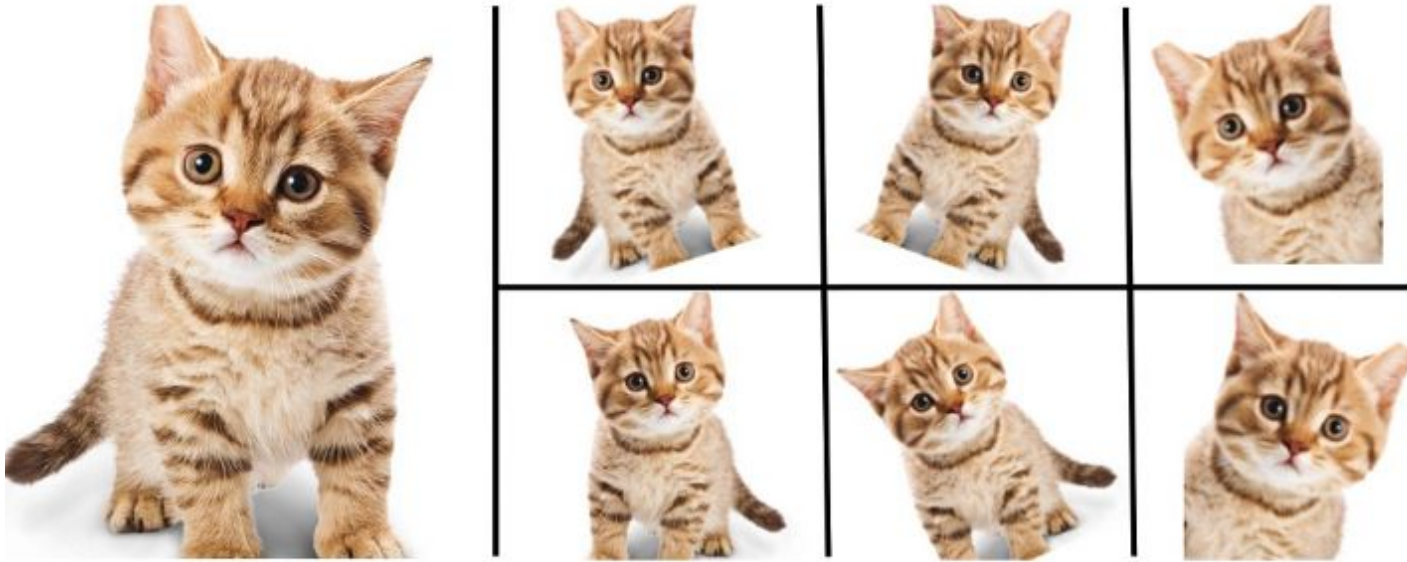
- Code
 - finetune_hf.ipynb
- Data
 - Set image dimensions & location
 - Read images from folder in batches
- Model
 - Load trained model from feature extraction code
 - Weights in last few convolutional blocks and top model will be adjusted during training
 - All other weights in pre-trained model are frozen
- Train model
 - Use training data to adjust top model weights
 - Use validation data to determine when to stop training
- Evaluate model
 - Calculate accuracy, etc.
 - Perform inference on test images

Early Stopping

Using validation data to determine when to stop training to avoid overfitting



Data Augmentation



Add variability to your dataset

<https://nanonets.com/blog/data-augmentation-how-to-use-deep-learning-when-you-have-limited-data-part-2/>

RESOURCES

- Transfer Learning
 - <http://cs231n.github.io/transfer-learning/>
- ImageNet
 - <http://www.image-net.org/>